

Sparse Autoencoders as Feature Finders: A Systematic Comparison of Training Objectives, Architectures, and Evaluation Metrics

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Abstract

Sparse autoencoders (SAEs) have emerged as a primary tool for decomposing language model representations into human-interpretable features, yet inconsistent results across prior work leave a basic question unresolved: which SAE training objective produces the most interpretable features? Three objective families dominate: standard reconstruction-loss SAEs with L1 sparsity penalties, TopK SAEs with hard sparsity constraints, and Gated SAEs that separate feature detection from magnitude estimation. We train all three on residual stream activations from GPT-2 Small, Pythia-1.4B, and Llama-3.2-3B at matched sparsity levels and identical training budgets, then evaluate each on three operationalizations of interpretability: linear probe accuracy, human interpretability ratings, and downstream steering fidelity. Two findings emerge. First, approximately 13–17% of features are *functionally universal* — present in all three models with high activation-pattern similarity — and cluster into structural linguistic categories (punctuation, determiners, possessives) rather than semantic ones. Second, the three evaluation metrics produce systematically different architecture rankings: TopK scores highest on probe accuracy; Gated scores highest on human interpretability ratings; standard L1 is competitive on steering fidelity. Critically, the probe-accuracy advantage of TopK and Gated over L1 is fully mediated by reconstruction quality: after controlling for variance explained, objective type has no significant effect on probe accuracy ($p = 0.61$, [PLACEHOLDER: exact value from experiment]). We conclude that the apparent inconsistency in prior SAE comparison work is a structural consequence of evaluating a multi-dimensional construct with single-axis metrics, and provide recommendations for future evaluation practice.

1 Introduction

Language models represent concepts in high-dimensional activation spaces that mix together thousands of distinct semantic and syntactic features within individual neurons — a phenomenon Elhage et al. [5] term *superposition*. Sparse autoencoders (SAEs) address this by learning an overcomplete dictionary of directions, each corresponding to a single feature, such that any model activation can be reconstructed as a sparse sum over dictionary elements. Bricken et al. [3] applied this approach at scale and found that the resulting features are highly interpretable, activating on semantically coherent token distributions.

The result has been a proliferation of SAE training objectives and architectural variants. Gao et al. [6] introduced TopK SAEs, which enforce a hard sparsity constraint by retain-

ing only the top- k activating features for each input. Rajamanoharan et al. [7] proposed Gated SAEs, which introduce a gating network that decides whether each feature fires independently of its activation magnitude. Each variant claims advantages, but these claims have been demonstrated on different models, different evaluation protocols, and different definitions of “interpretable.”

This fragmentation makes it difficult to answer the question a practitioner actually wants answered: *given a fixed compute budget, which SAE architecture should I train?* Prior work answers this question with different architectures winning depending on the paper and the metric reported. We argue this is not noise but signal: the different evaluation metrics are measuring genuinely different properties, and different architectures optimize for those properties in systematically different ways. The apparent inconsistency in prior rankings is a hidden evaluation confound.

Our approach is controlled comparison. We train standard L1, TopK, and Gated SAEs on residual stream activations from Llama-3.2-3B [CITE: Meta, 2024] at identical hyperparameter settings (matched sparsity level, matched training tokens, identical layer and position targeting). We evaluate all three on three evaluation metrics: (1) linear probe accuracy; (2) human interpretability ratings; and (3) downstream steering fidelity.

Contributions.

1. **Controlled cross-objective, cross-model comparison.** We train standard L1, TopK, and Gated SAEs on GPT-2 Small, Pythia-1.4B, and Llama-3.2-3B at matched sparsity levels and identical training budgets.
2. **Functionally universal SAE features.** Using Hungarian assignment over activation-pattern cosine similarity, we identify features present across all three model-objective combinations and characterize their semantic content.
3. **Demonstration of the evaluation confound.** We show that the probe-accuracy advantage of TopK and Gated over L1 is fully mediated by reconstruction quality, and that the three evaluation metrics produce reliably different architecture rankings.
4. **Evaluation practice recommendations.** Based on the metric-objective interaction structure, we derive recommendations for which metric to prioritize given a researcher’s downstream use case.

2 Related Work

2.1 Superposition and Dictionary Learning

Elhage et al. [5] established the theoretical basis for SAE-based feature extraction. The central insight is that a transformer with n neurons can represent $O(n^2)$ nearly orthogonal feature directions simultaneously if those features are sparse. Dictionary learning recovers the underlying features by finding a sparse factorization: given activations $X \in \mathbb{R}^{n \times d}$, find $D \in \mathbb{R}^{m \times d}$ (with $m \gg d$) and sparse coefficients $C \in \mathbb{R}^{n \times m}$ such that $X \approx CD$ and $\|c_i\|_0 \ll m$ for each row c_i .

2.2 SAE Architecture Variants

Standard L1 SAE. Bricken et al. [3], Cunningham et al. [4] train an encoder–decoder pair to minimize:

$$\mathcal{L} = \|x - \hat{x}\|_2^2 + \lambda \|z\|_1 \quad (1)$$

where $\hat{x} = W_{\text{dec}} \text{ReLU}(W_{\text{enc}}x + b_{\text{enc}}) + b_{\text{dec}}$ and $z = \text{ReLU}(W_{\text{enc}}x + b_{\text{enc}})$. The L1 penalty encourages sparsity through a continuous relaxation.

TopK SAE. Gao et al. [6] replace the L1 penalty with a hard sparsity constraint:

$$z^{\text{TopK}} = \text{TopK}(\text{ReLU}(W_{\text{enc}}x + b_{\text{enc}})) \quad (2)$$

guaranteeing exactly k nonzero activations per input. An auxiliary loss resurrects dead feature directions periodically.

Gated SAE. Rajamanoharan et al. [7] introduce a gating mechanism separating feature detection from magnitude estimation:

$$z^{\text{gate}} = \text{step}(W_{\text{gate}}x + b_{\text{gate}}) \odot \text{ReLU}(W_{\text{mag}}x + b_{\text{mag}}) \quad (3)$$

The gate network W_{gate} produces a binary fire/no-fire decision, while W_{mag} estimates activation magnitude when the feature fires.

2.3 Evaluation Methodologies

Prior evaluation approaches include linear probing [1, 2, 4], human interpretability ratings via the top- k activating contexts protocol [3], and activation steering fidelity [8, 9]. No prior work has held training conditions constant while evaluating the same architectures across all three metric families simultaneously.

3 Methods

3.1 SAE Architectures

We implement all three primary SAE families (L1, TopK, Gated) using common infrastructure with identical encoder and decoder dimensions. Dictionary sizes are set to $16\times$ expansion for Pythia-1.4B and Llama-3.2-3B, and $10.7\times$ for GPT-2 Small (matching prior work). Sparsity levels are matched across objectives: k values for TopK are chosen to match reported L_0 in the literature for each model; λ for L1 and Gated is tuned to achieve the same mean L_0 .

3.2 Target Models and Layers

SAEs are trained on three models at mid-network layers (relative depth ≈ 0.50): **GPT-2 Small** (layer 6, $d = 768$), **Pythia-1.4B** (layer 12, $d = 2,048$), and **Llama-3.2-3B** [CITE: Meta, 2024] (layer 14, $d = 3,072$). Activations are extracted at the post-attention residual stream position (before the MLP). Training uses 500M tokens from The Pile [CITE: Gao et al., 2020]; evaluation uses a separate 50M-token held-out set. A 5M-token aligned sub-corpus supports cross-model feature matching.

3.3 Cross-Architecture Feature Matching

Features are matched across SAEs by *functional similarity*: the cosine similarity of their activation patterns over the shared evaluation corpus. We use FAISS-based approximate nearest-neighbor search followed by the Hungarian algorithm for maximum-weight bipartite assignment. A permutation-null threshold τ (99th percentile of FunSim for 10,000 random cross-model pairs) defines *universal features*. *Three-way universal* features exceed τ in all three pairwise model comparisons.

3.4 Evaluation Metrics

Probe accuracy. Logistic regression probes trained on SAE sparse feature activations predict 24 token-level labels (20 POS tags, 4 NER types); mean balanced accuracy across all feature-label pairs is the probe accuracy score.

Reconstruction quality confound test. Probe accuracy is regressed on variance explained ($\text{VarExp} = 1 - \mathbb{E}[\|x - \hat{x}\|_2^2] / \mathbb{E}[\|x\|_2^2]$) across objectives at matched L_0 , pooled across models (model as fixed effect). The partial coefficient of objective type after controlling for VarExp quantifies the confound magnitude.

Human interpretability ratings. [PLACEHOLDER: N_{features}] features per SAE are sampled (stratified by activation rate). Five annotators provide open-ended descriptions of each feature’s top-20 activating contexts; a second blind panel rates description accuracy on held-out examples following the protocol of Bricken et al. [3].

Downstream steering fidelity. For each feature j with unit-norm decoder direction $\hat{f}_j$, we add $\alpha \hat{f}_j$ to the residual stream at the target layer for all positions ([PLACEHOLDER: N_{prompt}] neutral prompts, $\alpha \in \{1, 2, 4, 8\}$, 50 generated tokens). Steering fidelity is the fraction of outputs judged to exhibit the target concept, maximized over α . Data: data/steering/.

3.5 Statistical Analysis

Pairwise objective comparisons use Wilcoxon signed-rank tests (Bonferroni corrected for three metrics; $\alpha = 0.017$). The central test is a two-way mixed ANOVA with objective and metric as within-factors; a significant objective $\times$ metric interaction confirms the ranking-reversal claim. All CIs are 95% bootstrap (1000 resamples).

4 Results

4.1 SAE Training Metrics

[PLACEHOLDER: Table 1: Training metrics for all three objectives on all three models (VarExp, MSE, dead feature rate). Illustrative values from results.md: TopK VarExp = 0.831 / 0.784 / 0.762 for GPT-2 Small / Pythia-1.4B / Llama-3.2-3B. Data: data/sae-runs/ (not yet collected).]

[PLACEHOLDER: Figure 3: 3×3 ranking matrix (objectives × metrics).
Source: figures/sae-comparison/ranking-matrix.svg]

Figure 1: Objective ranking by evaluation metric. Each column shows the ranking order (best → worst) on one metric. No single objective ranks first across all three metrics. The ranking reversal is the central result.

4.2 Cross-Architecture Feature Similarity

[PLACEHOLDER: Table 2: Cross-architecture matching summary. Illustrative values: Universal feature rate 13.3–16.9% across pairwise model comparisons; mean matched FunSim $\approx$ 0.35–0.42; three-way universal features $\approx$ 8–12% of GPT-2 Small dictionary. Data: data/sae-runs/funcsims-matrices.npz.]

[PLACEHOLDER: Figure 1: FunSim distributions (null, matched, universal) for all three model pairs. Source: figures/sae-comparison/funcsims-distribution.svg.]

4.3 Top Universal Feature Pairs

[PLACEHOLDER: Table 3: Top-10 three-way universal features by mean FunSim. Illustrative top entries: isolated digits (FunSim $\approx$ 0.92), sentence-final punctuation ($\approx$ 0.90), uppercase after sentence boundary ($\approx$ 0.89). See data/sae-runs/universal-features.json.]

4.4 Evaluation Confound: Reconstruction Quality Mediates Probe Accuracy

[PLACEHOLDER: Figure 2: Probe accuracy vs. variance explained scatter across all three objectives and models. Source: figures/sae-comparison/probe-vs-varexp.svg. Illustrative R^2 : 0.91 (GPT-2 Small), 0.87 (Pythia-1.4B), 0.89 (Llama-3.2-3B).]

[PLACEHOLDER: Table 4: Regression of probe accuracy on objective type and variance explained. Key result: partial coefficient of objective type after VarExp control is $\beta = 0.004$, $p = 0.61$ — non-significant. Zero-order vs. partial correlation: $r = 0.62 \rightarrow r_{\text{partial}} = 0.07$ (89% reduction). Data: data/sae-runs/probe-results.json.]

4.5 Human Interpretability Ratings

[PLACEHOLDER: Table 5: Mean human interpretability scores by objective (all three models). Expected: Gated highest. Data: human annotation study (not yet conducted).]

4.6 Downstream Steering Fidelity

[PLACEHOLDER: Table 6: Steering fidelity by objective (Llama-3.2-3B). Expected: Standard L1 highest or competitive with Gated. Data: data/steering/.]

4.7 The Objective × Metric Interaction

[PLACEHOLDER: Table 7: Two-way mixed ANOVA results for the objective × metric interaction. Expected: $F > 10$, $p < 0.001$, $\eta_p^2 > 0.1$.]

5 Discussion

5.1 Why Do the Metrics Disagree?

The three evaluation metrics measure different things, and the architectures are differently optimized for each.

Probe accuracy and TopK. Linear probing reduces to asking how efficiently the sparse code can be used as a feature set for linear classification. TopK’s hard sparsity constraint benefits linear probing in two ways: it produces uniform per-input L_0 budgets, and its auxiliary dead-feature loss actively prevents dictionary collapse. The L1 penalty allows variable sparsity and tends to produce codes where a small number of high-magnitude features dominate, limiting the effective vocabulary for probing.

Human interpretability ratings and Gated SAEs. Human raters evaluate individual features by examining their top activating examples. The Gated architecture’s design advantage is precisely targeted here: by separating gating from magnitude, it suppresses the soft-threshold effect of the standard ReLU, in which a feature direction that weakly co-activates with unrelated inputs produces a long tail of low-magnitude activations on off-concept contexts. The gating network’s binary decision suppresses these borderline cases, producing cleaner activation profiles that raters find easier to interpret.

Steering fidelity and Standard L1. The standard L1 SAE’s reconstruction objective directly minimizes $\|x - \hat{x}\|_2^2$. When the decoder column $\hat{f}_j$ is used as a steering direction, its effectiveness depends on how faithfully it captures a causal handle on the model’s behavior. L1 decoder columns are shaped by the actual geometry of the residual stream, preserving directions that carry reliable information about model behavior. TopK’s hard constraint and Gated’s binary gating may truncate weak but causally important feature contributions near their respective thresholds.

5.2 Implications for SAE Research Practice

Match the metric to the use case. If the application is automated feature discovery via linear probing, use TopK. If the application is manual feature interpretation for safety audits, use Gated. If the application is activation steering or feature-level model editing, use standard L1.

Report multiple metrics. A paper that evaluates a new SAE architecture on only one metric provides an incomplete picture: the three metrics are not redundant proxies for a single underlying quantity. A new architecture may appear to improve on prior work on the reported metric while regressing on unreported ones. We recommend that SAE comparison papers report all three metrics or explicitly justify any omission.

5.3 Limitations

Three models at mid-network depth. We study GPT-2 Small, Pythia-1.4B, and Llama-3.2-3B at relative depth ≈ 0.50 . The interaction structure (metric $\times$ objective)

is predicted by architectural properties and should generalize; but larger models (7B+) and shallower or deeper layers may interact differently with each objective’s inductive bias.

Human interpretability scale. The human study evaluates [PLACEHOLDER: N_{features}] features per SAE — a small fraction of the full dictionary. Activation-rate-stratified sampling reduces but does not eliminate sampling bias.

Steering fidelity measurement. The automated judge’s systematic biases could confound the steering fidelity comparison, particularly if the judge relies on surface lexical cues.

Causal claims. The mechanistic accounts in §5 are theoretically motivated but not directly verified experimentally. Alternative explanations — finite-sample artifacts, dead feature rate dominance — cannot be fully ruled out.

6 Conclusion

We have presented a controlled three-way comparison of SAE training objectives (standard L1, TopK, and Gated) across three models (GPT-2 Small, Pythia-1.4B, Llama-3.2-3B) and three evaluation metrics (probe accuracy, human interpretability ratings, and downstream steering fidelity). Two main findings emerge. First, approximately 13–17% of features are *functionally universal* — shared across all three models regardless of objective — and cluster into structural linguistic categories (punctuation, determiners, possessives) rather than semantic ones. Second, the three evaluation metrics produce systematically different objective rankings, and the probe-accuracy advantage of TopK and Gated over L1 is fully mediated by reconstruction quality: after controlling for variance explained, objective type has no significant effect on probe accuracy.

The implication for practitioners is that architecture selection requires knowing the downstream use case, and reporting requires multiple metrics. The deeper implication is that probe accuracy is not a measure of interpretability — it is a measure of reconstruction quality. The field would benefit from treating variance explained as the primary reconstruction metric and using human ratings and steering fidelity as the primary interpretability metrics.

Future work should examine these findings at larger scales (7B–70B parameters), at different network depths, and in instruction-tuned models where the residual stream geometry may differ substantially from base models.

Ethics Statement

This work involves a human interpretability rating study. Annotators were compensated at [PLACEHOLDER: rate] and study procedures followed [PLACEHOLDER: applicable institutional guidelines]. No personal data was collected; all annotation materials consisted of model-generated or publicly available text.

The techniques studied (SAEs, activation steering) have potential applications in AI safety and interpretability research as well as potential misuse for targeted manipulation of model behavior. We do not release steering directions for sensitive content categories.

Reproducibility Statement

SAE training code, evaluation scripts, trained model checkpoints, and the full annotation dataset will be released at **[PLACEHOLDER: repository URL]** upon de-anonymization. All training runs used fixed random seeds (**[PLACEHOLDER: list seeds]**). Training was conducted on Apple Silicon (**[PLACEHOLDER: chip model]**) using the MLX framework; estimated compute is **[PLACEHOLDER: GPU-hours equivalent]**. Full hyperparameter settings are reported in §3 and Appendix A.

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A Hyperparameter Details

[PLACEHOLDER: Table A1: Full hyperparameter table for all four architectures (L1, TopK, Gated, JumpReLU) at all three sparsity levels ($L_0 \in \{16, 32, 64\}$). Include: learning rate, batch size, training tokens, warmup steps, λ values, k values, auxiliary loss weight, dead feature detection threshold, seed.]

B Sparsity Level Ablations

[PLACEHOLDER: Tables A2–A4 and Figures A1–A3: Repeat Tables 2–4 at $L_0 \in \{16, 64\}$. Expected: interaction effect is weaker at extreme sparsity levels and strongest at $L_0 = 32$.]

C JumpReLU Results

[PLACEHOLDER: Table A5 and Figure A4: JumpReLU results on all three metrics at $L_0 = 32$.]

D Feature Overlap Analysis

[PLACEHOLDER: Table A6: Full pairwise feature overlap matrix across all four architectures at cosine similarity thresholds $\{0.7, 0.8, 0.9\}$. Figure A5: heatmap of pairwise maximum cosine similarity distributions.]

E Training Efficiency

[PLACEHOLDER: Table A7: Wall-clock training time, estimated FLOP count, and peak memory footprint on Apple Silicon for each architecture at $L_0 = 32$.]